Regenerative Neuroscience Inspired Optimization Algorithm

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Abstract. Regenerative neuroscience is a cutting-edge field in neuroscience, focusing on the mechanisms of neural stem cell regeneration, differentiation, and neural network reconstruction following injury. Based on the biological mechanisms of regenerative neuroscience, this paper proposes a novel optimization algorithm, Regenerative Neuroscience Inspired Optimization (RNIO). This algorithm incorporates neural stem cell regeneration and differentiation, multipath exploration, a dynamic inhibitory environment, and synaptic network information integration mechanisms, fully simulating the self-repair process of the nervous system after injury. The algorithm's core iterative formula combines three mechanisms: regeneration and differentiation, local axon extension, and global synaptic integration, achieving a balance between global exploration and local exploitation. This paper deduces the algorithm's mathematical model and iterative formula in detail, analyzing the role and innovativeness of each mechanism, and provides a new theoretical approach for high-dimensional nonlinear optimization problems. This method is not only biologically inspired but also clearly interpretable from a theoretical perspective.

Keywords: Regenerative neuroscience; optimization algorithms; neural stem cells; multipathway differentiation; synaptic integration.

I. Introduction

Optimization algorithms have a wide range of applications in scientific research and engineering practice, including solving high-dimensional nonlinear problems, structural design optimization, energy system regulation, and machine learning hyperparameter tuning. In recent years, bioinspired algorithms have developed rapidly, such as particle swarm optimization (PSO), ant colony optimization (ACO), and genetic algorithms (GA). These algorithms are often inspired by animal behavior in nature. However, these algorithms often face problems such as premature convergence, local optimality, and insufficient exploration and exploitation balance in complex high-dimensional search spaces [1-46].

Regenerative neuroscience, as an important branch of neuroscience, studies the regeneration and differentiation of neural stem cells after injury and the functional recovery of neural networks. Its core features include:

Neural stem cell regeneration and differentiation: Neural stem cells have the ability to self-renew and differentiate into neurons, astrocytes, or oligodendrocytes, thereby repairing damaged tissues.

Axon growth and synaptic reconstruction: Regenerated neurons can extend axons along specific directions and form functional synapses with other neurons, achieving neural network reconstruction.

Inhibitory Environment Regulation: Injury to the central nervous system produces inhibitory molecules and glial scars, which restrict axon regeneration.

Based on these biological mechanisms, this paper proposes a novel optimization algorithm, the Regenerative Neuroscience-Inspired Optimization (RNIO). The algorithm incorporates multi-path differentiation, a dynamic inhibitory environment, and synaptic information integration mechanisms, achieving a combination of efficient global search and local refined search, fully embodying the characteristics of regenerative neuroscience.

II. Algorithm Design Concept

The RNIO algorithm design concept is entirely derived from the core mechanisms of regenerative neuroscience. Its main design principles are as follows:

Multi-path Differentiation Mechanism: This simulates the process by which neural stem cells differentiate into different neurons. Each candidate solution generates multiple differentiation paths during the iteration process, enabling diversified exploration of the search space.

Axon Extension Local Search: This simulates the process of axon growth. Candidate solutions are searched locally along the gradient of the objective function, while being controlled by a dynamic inhibitory environment to avoid over-clustering or premature convergence.

Synaptic Network Information Integration: This simulates the synaptic plasticity mechanism in neural networks. Candidate solutions exchange nonlinear weighted information between them, enabling global collaborative search.

Dynamic Inhibitory Environment: This algorithm simulates the regulatory effects of inhibitory molecules on neural regeneration, calculates the inhibitory factor based on neighborhood density and fitness difference, and adaptively adjusts the intensity of candidate solution exploration.

Through these four mechanisms, the RNIO algorithm achieves an innovative transition from biological inspiration to mathematical modeling.

III. Mathematical Model

3.1 Initial Population and Multi-Path Differentiation

Assume the search space of the optimization problem is D-dimensional and the initial population size is N. The initial candidate solutions are:

$X_i = [x_{i1}, x_{i2}, ..., x_{iD}]$, where $i = 1, 2, ..., N$

Each candidate solution can generate P differentiation paths during an iteration:

$$X_i^p = X_i + \lambda_p * (X_{best} - X_i) + \sigma_p * \varepsilon$$

Where:

X_i^p represents the p-th differentiation path of the i-th candidate solution

λ_p is the differentiation strength, which decreases with the path number

X_{best} is the current global optimal solution

σ_p is the random perturbation strength

ε is a Gaussian noise vector, simulating the randomness of the biological environment

This mechanism ensures that each candidate solution approaches the global optimum while maintaining search diversity, simulating the multi-directional differentiation ability of neural stem cells.

3.2 Dynamic Inhibition Factor

Each differentiation path is inhibited by the neighborhood environment. Its inhibition factor, I_i^p, is determined by the density of the neighborhood solution and the fitness gradient:

$$I_i^p = \delta * \Sigma_{j \in Neigh(i)} \exp(-||X_i^p - X_j||^2 / \sigma_i^2) * (f(X_i^p) - f(X_j))$$

Where:

δ is the inhibition coefficient

Neigh(i) is the neighborhood set of the i-th candidate solution

σ_i is the local scaling factor

$f(X)$ is the objective function value

The dynamic inhibition factor effectively prevents differentiation paths from over-aggregating in high-density areas, achieving adaptive control of the search space.

3.3 Axon Extension Mechanism (Local Exploration)

Local search is implemented through the axon extension mechanism, which is regulated by an inhibitory factor:

$$\Delta X_i^p = \alpha * grad_f(X_i^p) * (1 - I_i^p)$$

$grad_f(X_i^p)$ is the objective function gradient

α is the local exploration intensity coefficient

The candidate solution update formula is:

$$X_i^p = X_i^p - \Delta X_i^p$$

This mechanism simulates the process of a neuron extending its axon along the gradient direction, while being regulated by an inhibitory environment.

3.4 Synaptic Integration Mechanism (Global Coordination)

Information integration between candidate solutions is performed via a nonlinear synaptic network, as follows:

$$X_i^p = X_i^p + \gamma * \Sigma_{j=1}^N \Sigma_{q=1}^P w_{ij}^{pq} * (X_j^q - X_i^p)$$

Where the weight w_{ij}^{pq} is determined by the distance and fitness difference:

$$w_{ij}^{pq} = \exp(-\|X_i^p - X_j^q\|^2 / \tau) * S(f(X_j^q) - f(X_i^p))$$

The fitness gain function $S(x)$ is defined as:

$$S(x) = 1 / (1 + \exp(-k * x))$$

γ is the global integration coefficient

τ is the distance scale parameter

k is the S Function Slope Coefficient

This mechanism simulates synaptic plasticity in neural networks, achieving information self-organization and global coordination.

3.5 Comprehensive Iterative Formula

Combining regeneration and differentiation, axon extension, and synaptic integration, the iterative formula for the RNIO algorithm is:

$$
\begin{aligned}
X_i^p(t+1) &= X_i^p(t) \\
&+ \lambda_p * (X_{best} - X_i^p(t)) + \sigma_p * \varepsilon \\
&- \alpha * grad_f(X_i^p(t)) * (1 - I_i^p) \\
&+ \gamma * \Sigma_{j=1}^N \Sigma_{q=1}^P w_{ij}^{pq} * (X_j^q(t) - X_i^p(t))
\end{aligned}
$$

This formula embodies the algorithm's multi-layered mechanisms: global guidance, local refined search, and information integration.

IV. Algorithm Flow

The main flow of the RNIO algorithm is as follows:

Input: Objective function f(X), population size N, maximum number of iterations T, number of differentiation paths per solution P
Output: Optimal solution X_best

1. Initialize population X_i (i=1,...,N)
2. Evaluate fitness f(X_i)
3. Set X_best = argmin(f(X_i))

4. For t = 1 to T:
4.1 For i = 1 to N:
4.1.1 For p = 1 to P:
Regenerative differentiation
$X_i^p = X_i + \lambda_p*(X_{best} - X_i) + \sigma_p*Gaussian_noise$

Dynamic suppression factor
$I_i^p = \delta * \Sigma_{j \in Neigh(i)} \exp(-||X_i^p - X_j||^2 / \sigma_i^2) * (f(X_i^p) - f(X_j))$

Axon extension
$\Delta X_i^p = \alpha * grad_f(X_i^p) * (1 - I_i^p)$
$X_i^p = X_i^p - \Delta X_i^p$

Synaptic integration
$X_i^p = X_i^p + \gamma * \Sigma_{j=1}^N \Sigma_{q=1}^P w_{ij}^{pq} * (X_j^q - X_i^p)$

4.2 Update $X_best = argmin(f(X_i^p))$

5. Return X_best

V. Mechanism Analysis and Innovations

Multi-path differentiation mechanism: Simulates the multidirectional differentiation of neural stem cells to ensure search diversity and global coverage.

Dynamic Inhibition Environment: Automatically adjusts local search intensity based on neighborhood density and fitness differentials to avoid premature convergence.

Local Exploration of Axon Extension: Combines gradient information and inhibitory factors to achieve refined local search, simulating the axon growth process of neurons.

Global Synaptic Integration: A nonlinear weighted mechanism simulates synaptic plasticity, enabling efficient information transmission between populations.

Exploration-Exploitation Balance: Balances global exploration and local exploitation through multi-path differentiation and synaptic integration mechanisms, theoretically adapting to complex high-dimensional optimization problems.

Mathematical Interpretability: All mechanisms are clearly described mathematically, and the formulas directly correspond to biological mechanisms, ensuring theoretical clarity of the algorithm.

VI. Potential Applications and Prospects

The RNIO algorithm has a wide range of potential applications:

High-dimensional nonlinear optimization problems: Suitable for optimizing complex functions in continuous or discrete search spaces.

Engineering Design Optimization: Suitable for structural optimization, energy system regulation, network design, and other problems.

Machine Learning Hyperparameter Optimization: Global search and local fine-tuning of deep learning model training parameters.

Complex System Simulation and Control: Multi-objective Optimization, Resource Scheduling, and Complex System Regulation.

Future research could combine neural engineering simulation with artificial intelligence technologies to simulate and verify algorithm performance, and further expand this approach to multi-objective optimization and multi-agent collaborative optimization.

VII. Conclusion

This paper proposes a novel optimization algorithm based on regenerative neuroscience — RNIO. Through multi-path differentiation, a dynamic inhibitory environment, axon extension, and synaptic integration mechanisms, the algorithm efficiently combines global search with local exploitation. The algorithm design fully reflects the biological characteristics of regenerative neuroscience and is rigorously derived through mathematical models, resulting in formulas that clearly correspond to biological mechanisms. The RNIO algorithm is not only theoretically original and interpretable, but also provides a new approach and method for complex high-dimensional optimization problems. Future research will explore its potential in engineering optimization and artificial intelligence by combining simulation with practical applications.

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